Week 11: 09 – 13 November 2026

Subject: Mathematics **Unit Title:** Statistics **Grade :** 12DP

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1 Grouped Data: Estimated Mean & Cumulative Frequency Graph	• Construct frequency tables for grouped data. • Estimate the **mean** for grouped data using the midpoint method. • Determine the **modal class** using the highest frequency. • Draw and interpret a **cumulative frequency graph** to find median and quartiles.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 8.1 Make goals clear, visible, and supported by short-term objectives.	Reflection prompts, notebooks The resources will be available on student's Teams channel Revision Village problems based on variations (direct and indirect) and geometry of 3D shapes. https://www.revisionvillage.com/ Past IB AI SL questions on Statistics. https://www.savemyexams.com/members/ InThinking online assessment to measure student progress on the concept understanding of Statistics. https://thinkib.net/mathapplications
Lesson 2 Cumulative Frequency Practice and Analysis	• Interpret cumulative frequency graphs to determine **median, quartiles, and interquartile range (IQR)**. • Compare two data sets using IQR and spread. • Use cumulative frequency to analyze **data distribution and skewness**.		
Lesson 3 Correlation (Pearson's and Spearman's)	• Explain the concept of **correlation** between variables. • Compute **Pearson's product-moment correlation coefficient** using calculator or software. • Calculate **Spearman's rank correlation coefficient** and interpret its meaning. • Distinguish between **causation and correlation** in real-world contexts.		
Lesson 4 Scatter Diagram, Linear Regression & Line of Best Fit	• Construct a **scatter diagram** for bivariate data. • Use technology to determine the **line of best fit (y = a + bx)**. • Interpret the **slope and intercept** in context. • Predict outcomes using regression equations.		

Teacher's notes/ Reminders ملاحظات المعلم والتذكير	

Technology Integration دمج التكنولوجيا	Use Desmos, GeoGebra and Excel for data visualization, boxplots and frequency tables; Teams for sharing and reflection.
Cross-curricular links الروابط بين المواد الدراسية	Science (data analysis of experiments), Economics (market sampling), Geography (population data comparison).
External Assessment Integration دمج التقييمات الخارجية	All content is covered in IB Past Exams
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Wellbeing:** • Confidence in solving real-world maths problems. • Resilience when learning from mistakes. • Collaboration through shared investigations and discussions. • Self-awareness by setting and reflecting on learning goals. **IB ATL Skills:** • **Thinking Skills** – Interpret numerical data to identify patterns and trends. • **Communication Skills** – Present and explain statistical findings clearly. • **Self-Management Skills** – Plan and execute data collection and organization. • **Research Skills** – Use authentic data sources to analyze results. • **Collaboration Skills** – Work in groups to design surveys and interpret results.
Standards المعايير	IB Math AI SL Standards: • 7.SP.A.1-2 • 7.SP.B.3 • HSS-ID.A.1-2 • 8.EE.B.5-6 • 8.F.B.4 • HSA-CED.A.2 • HSF-BF.A.1

Inquiry/ TOK Question **البحث والاستقصاء أسئلة** **TOK**	How can we trust data to tell the truth if the method of collecting it is biased?
Success Criteria معايير النجاح (Measurable)	Lesson 1: ▨ **All Students Must:** construct a frequency table for grouped data and identify the modal class. ◍ **Most Students Should:** estimate the mean of grouped data using class midpoints and frequencies. ▲ **Some Students Could:** draw and interpret a cumulative frequency graph to determine the median and quartiles. Lesson 2: ▨ **All Students Must:** find the median and quartiles from a cumulative frequency graph. ◍ **Most Students Should:** calculate and compare the interquartile range (IQR) of two data sets. ▲ **Some Students Could:** analyse the shape of a distribution and describe its spread and skewness using cumulative frequency graphs. Lesson 3: ▨ **All Students Must:** identify and describe the strength and direction of correlation between two variables. ◍ **Most Students Should:** calculate and interpret Pearson's and Spearman's correlation coefficients. ▲ **Some Students Could:** evaluate real-world data by explaining the difference between correlation and causation. solve questions involving algebra, functions, geometry & trigonometry, and statistics & probability. Lesson 4: ▨ **All Students Must:** construct and interpret a scatter diagram for bivariate data. ◍ **Most Students Should:** use technology to find and interpret a line of best fit, including its slope and intercept. ▲ **Some Students Could:** use a regression equation to make predictions and assess the reasonableness of those predictions in context. solve questions involving algebra, functions, geometry & trigonometry, and statistics & probability.
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